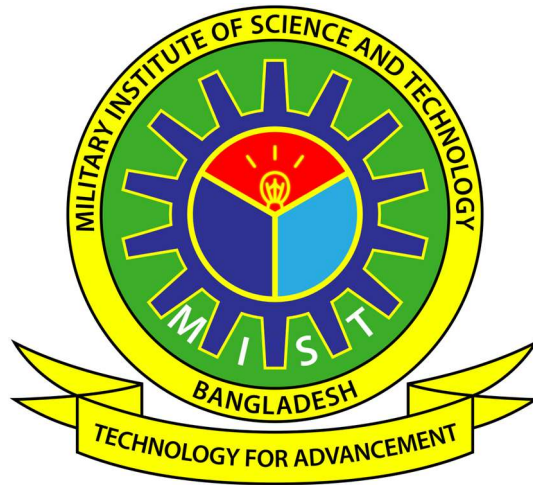




Term Paper

CSE-405

Computer Interfacing

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1. Daisy-Chain Interrupt Priority Scheme

A CPU has five interrupt lines: INTR1, INTR2, INTR3, INTR4, and INTR5, where INTR1 has the highest priority and INTR5 has the lowest. These lines are connected using a daisy-chain priority scheme.

Scenario:

- INTR3 is currently being serviced.
- INTR1 becomes active during this time.

When Nested Interrupts Are Enabled

If nested interrupts are enabled, the CPU allows higher-priority interrupts to interrupt the current service routine. Since INTR1 has a higher priority than INTR3, the CPU will:

1. Acknowledge INTR1 while servicing INTR3.
2. Suspend the current INTR3 ISR.
3. Push the context to the stack.
4. Service INTR1.
5. Resume INTR3 after INTR1 completes.

Result: INTR1 is serviced immediately, even though INTR3 is active.

When Nested Interrupts Are Disabled

If nested interrupts are disabled, the CPU will not acknowledge any new interrupt while one is being serviced. Therefore, the CPU continues to service INTR3 without interruption. INTR1 remains pending and will be serviced only after INTR3's ISR finishes.

Result: INTR1 must wait until INTR3's service routine finishes.

2. Programming the Intel 8237 DMA Controller for Audio Playback

Given:

- Audio samples stored at 4000H–47FFH (2 KB memory).
- Sound card buffer mapped at 5000H–57FFH.
- Channel 2 is used as the source (memory), and Channel 3 as the destination (I/O buffer).
- Transfer mode: Single-Transfer Mode.

The Intel 8237 DMA controller can transfer data between memory and I/O devices without CPU intervention. Below is the procedure to program the controller:

Step 1: Mask DMA Channels

OUT 0x0A, 82H ; Mask Channel 2
OUT 0x0A, 83H ; Mask Channel 3

Step 2: Set DMA Address Registers

Channel 2 (4000H):

OUT 0x04, 00H

OUT 0x04, 40H

Channel 3 (5000H):

OUT 0x06, 00H

OUT 0x06, 50H

Step 3: Set DMA Count Registers

2048 bytes → count = 2047 = 07FFH

Channel 2:

OUT 0x05, FFH

OUT 0x05, 07H

Channel 3:

OUT 0x07, FFH

OUT 0x07, 07H

Step 4: Set Mode Register

OUT 0x0B, 46H ; Channel 2, single mode, read transfer

OUT 0x0B, 4FH ; Channel 3, single mode, write transfer

Step 5: Unmask DMA Channels

OUT 0x0A, 02H ; Unmask Channel 2

OUT 0x0A, 03H ; Unmask Channel 3

Step 6: Start DMA Transfer

DMA controller now transfers data between 4000H–47FFH and 5000H–57FFH automatically.

3. Stepper Motor Interfacing with 8086 and 82C55 (Half-Step Mode)

A packaging factory uses a stepper motor for controlling a conveyor belt. For smoother motion and higher resolution, the motor is operated in half-step mode. The stepper motor is controlled using the Intel 82C55 PPI, which connects to the 8086 microprocessor. The motor windings are connected through driver circuits to the output pins of the 82C55.

i) Principle of Half-Step Operation

In half-step mode, the stepper motor alternates between one-phase ON and two-phase ON states. This doubles the resolution and reduces jerks. The clockwise excitation sequence is:

Step	A	B	C	D
1	1	0	0	0
2	1	1	0	0
3	0	1	0	0
4	0	1	1	0
5	0	0	1	0
6	0	0	1	1
7	0	0	0	1
8	1	0	0	1

The 8086 connects to the 82C55, which drives the stepper motor windings (A–D) through a driver circuit.

ii) Assembly Procedure for One Full Revolution

; Rotate Stepper Motor One Full Revolution in Half-Step Mode

MOV SI, OFFSET STEP_SEQ

MOV CX, 8

NEXT_STEP:

MOV AL, [SI]

OUT PORTA, AL

CALL DELAY

INC SI

LOOP NEXT_STEP

STEP_SEQ DB 00000001B, 00000011B, 00000010B, 00000110B

DB 00000100B, 00001100B, 00001000B, 00001001B
